

IIT Delhi MISN Lab Internship — Week 1 Assignment
The XOR Problem

Objective -:To understand linear separability using AND, OR, and XOR logic gates by plotting their truth table outputs on a 2D graph and analyzing whether a single perceptron can separate the outputs using a straight-line decision boundary.

★ **Solution** -:

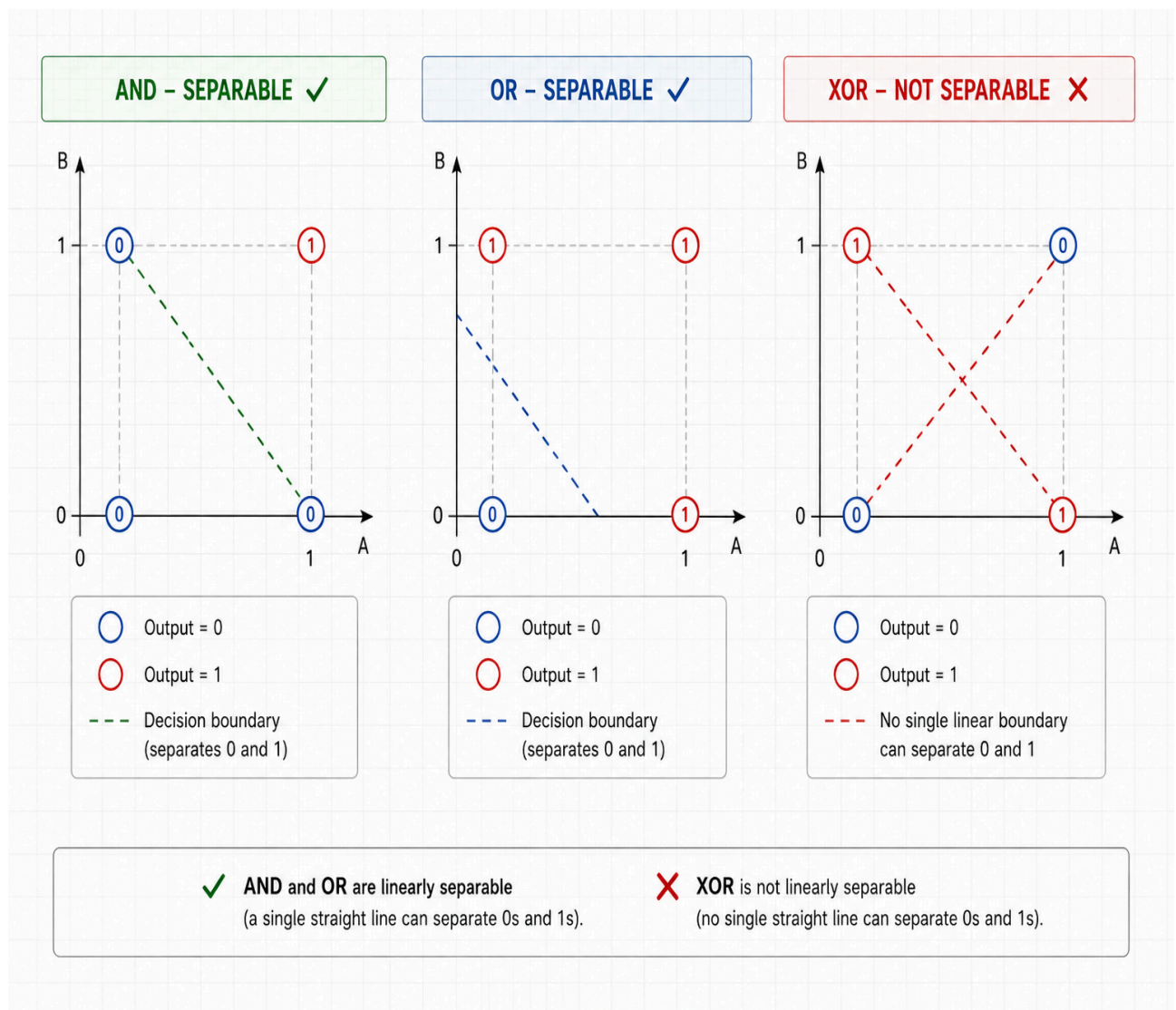
❖ **Truth Tables** -:

A	B	A And B
0	0	0
0	1	0
1	0	0
1	1	1

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

❖ 2-D Geometric View -:



Understanding Linear Separability -:

A single-layer perceptron is, at its core, a linear classifier. It computes a weighted sum of its inputs, passes it through a step function, and outputs 0 or 1. Geometrically, this is identical to drawing a straight line (or a hyperplane in higher dimensions) that divides the input space into two regions — one for each class. For AND, three of the four input points output 0 and only (1,1) outputs 1. A diagonal line cleanly places (1,1) on one side and the other three on the other — linearly separable. For OR, only (0,0) outputs 0 — again, a single line cuts it off from the three 1s with ease.

Why XOR Cannot Be Solved by a Single Perceptron -:

The XOR gate produces an output of 1 when the two inputs are different, such as (0,1) and (1,0), and produces 0 when the inputs are the same, such as (0,0) and (1,1). When these points are plotted on a 2D graph, the output 1 points lie diagonally opposite to each other, and the output 0 points also lie diagonally opposite. Because of this arrangement, no single straight line can separate all the 1s from the 0s correctly.

This makes XOR a non-linearly separable problem. Since a single-layer perceptron can create only one linear decision boundary, it cannot solve the XOR problem. This limitation led to the development of multi-layer perceptrons and neural networks, which can learn more complex decision boundaries.

Key Insight:

XOR cannot be solved by a single perceptron because no linear decision boundary exists for it. The solution is to stack layers — a hidden layer learns intermediate representations (essentially computing AND and OR internally), and the output layer combines them. This is exactly why the multi-layer perceptron (MLP) was invented, and XOR remains its canonical

"Hello World." This made me realise why a single neuron is fundamentally limited — it can only ask one yes/no question about the input space.